MATHS MANTHAN

• TUITION CLASSES •

e-mail: contact@mathsmanthan.co.in | Website:- mathsmanthan.co.in

Subject: Mathematics **WEEKLY CHAPTER TEST – 05 (SOLUTIONS)** **Time:** 45 Minutes
Class: Class X (CBSE) **Chapter 6: Triangles** **MM:** 20

Document Type: Official Step-by-Step Marking Scheme **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit theorem/formula declaration, neat geometric diagrams, logical reasoning, and final boxed results.
2. In descriptive proofs, state **Given**, **To Prove**, **Construction** (where needed), and justify every equality or similarity criterion.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 1.5M, 2M]**. Alternative valid methods receive full credit.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice Questions

MATHS MANTHAN | (5 × 1 = 5 Marks)

Q1. If two similar triangles have scale factor $a : b$, then the ratio of their corresponding altitudes is: [1]

(a) $a^2 : b^2$ (b) $a : b$ (c) $\sqrt{a} : \sqrt{b}$ (d) $2a : b$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (b) $a : b$

We know the ratio property of linear elements in similar triangles:

Formula / Theorem Used: If two triangles are similar with side ratio (scale factor) $a : b$, then the ratio of their corresponding altitudes, medians, angle bisectors, and perimeters is also $a : b$.

If $\triangle ABC \sim \triangle DEF$ with $\frac{AB}{DE} = \frac{a}{b}$, and $AL \perp BC$, $DM \perp EF$, then $\triangle ABL \sim \triangle DEM$ (AA similarity: $\angle B = \angle E$ and $\angle ALB = \angle DME = 90°$), giving $\frac{AL}{DM} = \frac{AB}{DE} = \boxed{\mathbf{a : b}}$.

Q2. If two similar triangles have scale factor 3 : 4 and the perimeter of the smaller triangle is 36 cm, the perimeter of the larger triangle is: [1]

(a) 48 cm (b) 45 cm (c) 54 cm (d) 42 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (a) 48 cm

Given: Scale factor = 3 : 4, perimeter of smaller triangle = 36 cm. Let perimeter of larger triangle be P cm.

We know the perimeter ratio theorem for similar triangles:

Formula / Theorem Used: $\dfrac{\text{Perimeter of Smaller Triangle}}{\text{Perimeter of Larger Triangle}} = \text{Scale Factor}$

Substituting the values:

$$\frac{36}{P} = \frac{3}{4} \implies 3P = 36 \times 4 = 144 \implies P = \frac{144}{3} = \boxed{\mathbf{48}\ \text{cm}}$$

Q3. In $\triangle ABC$, D and E are midpoints of sides AB and AC respectively. Then the ratio of DE to BC is: [1]

(a) 1 : 1 (b) 1 : 2 (c) 1 : 4 (d) 2 : 1

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (b) 1 : 2

Given: D and E are the midpoints of AB and AC respectively ($\frac{AD}{AB} = \frac{AE}{AC} = \frac{1}{2}$).
We know the Midpoint Theorem (and SAS similarity on $\triangle ADE \sim \triangle ABC$):

Formula / Theorem Used: The segment joining the midpoints of two sides of a triangle is parallel to the third side and half as long ($DE = \frac{1}{2}BC$).

Hence, $\frac{DE}{BC} = \frac{1}{2} \implies \boxed{\mathbf{1:2}}$.

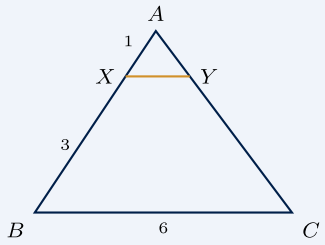
Q4. In the given figure, $XY \parallel BC$ in $\triangle ABC$. If $AX = 1$ cm, $XB = 3$ cm, and $BC = 6$ cm, then XY is: [1]

(a) 1.5 cm (b) 2 cm (c) 1 cm (d) 2.5 cm

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (a) 1.5 cm

Given: $XY \parallel BC$, $AX = 1$ cm, $XB = 3$ cm, $BC = 6$ cm.
Total side length $AB = AX + XB = 1 + 3 = 4$ cm.
Since $XY \parallel BC$, $\angle AXY = \angle B$ and $\angle A$ is common, so $\triangle AXY \sim \triangle ABC$ (AA similarity):



Formula / Theorem Used: $\frac{XY}{BC} = \frac{AX}{AB}$

Substituting the values:

$$\frac{XY}{6} = \frac{1}{4} \implies XY = \frac{6}{4} = \frac{3}{2} = \boxed{\mathbf{1.5}\text{ cm}}$$

Q5. **Assertion (A):** If $\triangle ABC \sim \triangle DEF$ such that $\frac{AB}{DE} = \frac{2}{3}$ and perimeter of $\triangle ABC$ is 16 cm, then perimeter of $\triangle DEF$ is 24 cm.
Reason (R): The ratio of the perimeters of two similar triangles is equal to the ratio of their corresponding sides. [1]

(a) Both (A) and (R) are true and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(c) Assertion (A) is true but Reason (R) is false.
(d) Assertion (A) is false but Reason (R) is true.

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (a) Both (A) and (R) are true and (R) is the correct explanation of (A).

We know the theorem stated in Reason (R):

Formula / Theorem Used: $\triangle ABC \sim \triangle DEF \implies \frac{\text{Perimeter}(\triangle ABC)}{\text{Perimeter}(\triangle DEF)} = \frac{AB}{DE}$

Substituting $\frac{AB}{DE} = \frac{2}{3}$ and $\text{Perimeter}(\triangle ABC) = 16$ cm:

$$\frac{16}{\text{Perimeter}(\triangle DEF)} = \frac{2}{3} \implies \text{Perimeter}(\triangle DEF) = \frac{16 \times 3}{2} = 8 \times 3 = 24 \text{ cm}$$

Hence, both (A) and (R) are true and (R) is the correct explanation of (A).

Q6. If $\triangle ABC \sim \triangle DEF$ such that $AB = 5$ cm, $DE = 7.5$ cm, and the perimeter of $\triangle DEF$ is 36 cm, calculate the perimeter of $\triangle ABC$. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Formula Declaration & Scale Factor [1 Mark]

Given: $\triangle ABC \sim \triangle DEF$, $AB = 5$ cm, $DE = 7.5$ cm, and Perimeter$(\triangle DEF) = 36$ cm.

We know the perimeter property of similar triangles:

Formula / Theorem Used: $\dfrac{\text{Perimeter}(\triangle ABC)}{\text{Perimeter}(\triangle DEF)} = \dfrac{AB}{DE}$

Step 2: Substitution & Calculation [1 Mark]

Substituting the given values into the ratio:

$$\frac{\text{Perimeter}(\triangle ABC)}{36} = \frac{5}{7.5} = \frac{50}{75} = \frac{2}{3}$$

Multiplying both sides by 36:

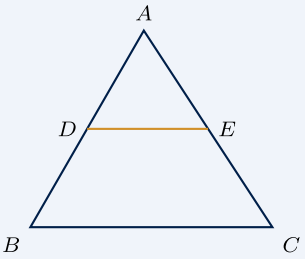
$$\text{Perimeter}(\triangle ABC) = \frac{2}{3} \times 36 = 2 \times 12 = 24 \text{ cm} \implies \boxed{\text{Perimeter}(\triangle ABC) = \mathbf{24} \text{ cm}}$$

Q7. Prove that the line segment joining the midpoints of any two sides of a triangle is parallel to the third side and equal to half of it. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: In $\triangle ABC$, D is the midpoint of AB $(AD = DB)$ and E is the midpoint of AC $(AE = EC)$.

To Prove: (i) $DE \parallel BC$, and (ii) $DE = \dfrac{1}{2}BC$.



We know the Converse of Basic Proportionality Theorem and AA similarity:

Formula / Theorem Used: If $\dfrac{AD}{DB} = \dfrac{AE}{EC}$, then $DE \parallel BC$; and similar triangles have proportional sides.

Step 1: Proving $DE \parallel BC$ [1.5 Marks]

Since D is the midpoint of AB and E is the midpoint of AC:

$$AD = DB \implies \frac{AD}{DB} = 1 \quad \text{and} \quad AE = EC \implies \frac{AE}{EC} = 1$$

Thus, $\dfrac{AD}{DB} = \dfrac{AE}{EC} = 1$. By the **Converse of BPT**, $DE \parallel BC$.

Step 2: Proving $DE = \dfrac{1}{2}BC$ [1.5 Marks]

In $\triangle ADE$ and $\triangle ABC$: $\angle ADE = \angle B$ (corresponding angles, as $DE \parallel BC$) and $\angle A = \angle A$ (common). Thus, $\triangle ADE \sim \triangle ABC$ (AA similarity).

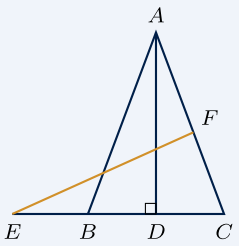
Since $AB = AD + DB = AD + AD = 2AD$:

$$\frac{DE}{BC} = \frac{AD}{AB} = \frac{AD}{2AD} = \frac{1}{2}$$

$$\boxed{\therefore DE \parallel BC \text{ and } DE = \frac{1}{2}BC \quad \text{[Hence Proved]}}$$

Q8. In the given figure, E is a point on side CB produced of an isosceles triangle ABC with $AB = AC$. If $AD \perp BC$ and $EF \perp AC$, prove that $\triangle ABD \sim \triangle ECF$. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]



Given: $\triangle ABC$ is isosceles with $AB = AC$; E lies on CB produced; $AD \perp BC$ and $EF \perp AC$.
To Prove: $\triangle ABD \sim \triangle ECF$.

We know the isosceles triangle angle theorem and AA similarity criterion:

Formula / Theorem Used: Angles opposite equal sides of a triangle are equal. Two triangles with two pairs of equal angles are similar (AA).

Step 1: Equal Angles from Isosceles Property [1 Mark]
In $\triangle ABC$, since $AB = AC$, the angles opposite to them are equal:

$$\angle ABC = \angle ACB \implies \angle ABD = \angle ECF \quad \text{— (i)}$$

Step 2: Comparing $\triangle ABD$ and $\triangle ECF$ [2 Marks]
In $\triangle ABD$ and $\triangle ECF$:

1. $\angle ABD = \angle ECF$ (from (i))
2. $\angle ADB = \angle EFC = 90^\circ$ (since $AD \perp BC$ and $EF \perp AC$)

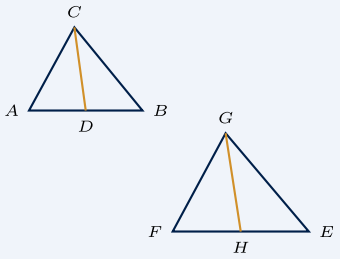
By the **AA similarity criterion**:

$$\therefore \triangle ABD \sim \triangle ECF \quad \text{[Hence Proved]}$$

MATHS MANTHAN • TUITION CLASSES •
MATHS MANTHAN OFFICIAL SOLUTIONS

Q9. In $\triangle ABC$ and $\triangle EFG$, CD and GH are respectively the medians to sides AB and FE. If $\triangle ABC \sim \triangle FEG$, prove that: (i) $\triangle ADC \sim \triangle FHG$, (ii) $\dfrac{CD}{GH} = \dfrac{AC}{FG}$. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]



Given: $\triangle ABC \sim \triangle FEG$; CD is median to AB ($AB = 2AD$); GH is median to FE ($FE = 2FH$).

To Prove: (i) $\triangle ADC \sim \triangle FHG$, (ii) $\dfrac{CD}{GH} = \dfrac{AC}{FG}$.

We know the definition of similarity and the SAS similarity criterion:

Formula / Theorem Used: If $\triangle ABC \sim \triangle FEG$, then $\angle A = \angle F$ and $\dfrac{AB}{FE} = \dfrac{BC}{EG} = \dfrac{AC}{FG}$.

Step 1: Proving $\triangle ADC \sim \triangle FHG$ [2 Marks]

From $\triangle ABC \sim \triangle FEG$, we have:

$$\angle A = \angle F \qquad \text{and} \qquad \frac{AB}{FE} = \frac{AC}{FG}$$

Since D and H are midpoints of AB and FE, substitute $AB = 2AD$ and $FE = 2FH$:

$$\frac{2AD}{2FH} = \frac{AC}{FG} \implies \frac{AD}{FH} = \frac{AC}{FG}$$

In $\triangle ADC$ and $\triangle FHG$, $\dfrac{AD}{FH} = \dfrac{AC}{FG}$ and included angle $\angle A = \angle F$.

By the **SAS similarity criterion**, $\triangle$**ADC** $\sim \triangle$**FHG**.

Step 2: Deducing Median Ratio [1 Mark]

Since corresponding sides of similar triangles $\triangle ADC$ and $\triangle FHG$ are proportional:

$$\therefore \frac{CD}{GH} = \frac{AC}{FG} \quad \text{[Hence Proved]}$$

MATHS MANTHAN • TUITION CLASSES • OFFICIAL SOLUTIONS

Q10. Case Study 05: Forestry & Tree Height Estimation Using Similar Triangles

A forestry officer needs to measure the height of a mature redwood tree without climbing it. She places a vertical measuring pole of height 2.5 m at a distance of 36 m from the base of the tree. Walking back an additional 4 m along the line of sight, she aligns her eye with the top of the pole and the top of the tree. Her eye level is 1.5 m above the ground. [4]

(i) **[1 Mark]** What is the height of the measuring pole above her eye level?

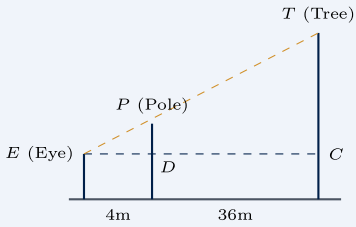
(ii) **[1 Mark]** What is the total horizontal distance from her eye to the base of the tree?

(iii) **[2 Marks]** Calculate the total vertical height of the redwood tree above the ground.

▶ **CASE-BASED SOLUTION & MARKING SCHEME:** [Total: 4 Marks]

Given Data: Eye level $EA' = 1.5$ m; total pole height $PB' = 2.5$ m; horizontal distance from eye to pole $= 4$ m; horizontal distance from pole to tree $= 36$ m.

Above the horizontal eye-level line EC, the vertical pole (PD) and vertical tree (TC) form two right triangles ($\triangle EPD$ and $\triangle ETC$) sharing $\angle E$ and having $\angle EDP = \angle ECT = 90°$, so $\triangle EPD \sim \triangle ETC$ (AA similarity):



Formula / Theorem Used: $\dfrac{\text{Tree height above eye level } (TC)}{\text{Pole height above eye level } (PD)} = \dfrac{\text{Eye-to-tree distance } (EC)}{\text{Eye-to-pole distance } (ED)}$

Sub-part (i): Height of Measuring Pole Above Eye Level [1 Mark]

$PD = \text{Total pole height} - \text{Eye level height} = 2.5 - 1.5 = 1.0 \text{ m} \implies$ Pole height above eye $= \mathbf{1.0}$ m

Sub-part (ii): Total Horizontal Distance from Eye to Tree Base [1 Mark]

$EC = \text{Eye-to-pole distance} + \text{Pole-to-tree distance} = 4 + 36 = 40 \text{ m} \implies$ Total distance $= \mathbf{40}$ m

Sub-part (iii): Total Vertical Height of the Redwood Tree Above Ground [2 Marks]

Let $TC = h'$ m be the height of the tree above eye level. By AA similarity ($\triangle ETC \sim \triangle EPD$):

$$\frac{h'}{1.0} = \frac{40}{4} = 10 \implies h' = 10 \times 1.0 = 10 \text{ m}$$

Adding the eye level height (1.5 m) above the ground:

$$\text{Total height of tree} = h' + 1.5 = 10 + 1.5 = 11.5 \text{ m} \implies \text{Total Tree Height} = \mathbf{11.5} \text{ m}$$

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★